Darshit Sharma Data Scientist | Machine Learning Engineer | AI Engineer

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WORK EXPERIENCE

Data Scientist- Mahindra Group Aug 2024 – present

AI Digital Twin of Wheel Force Transducer

- Developed a multi-output regression model using Temporal Convolutional Networks and XGBoost to predict wheel-end forces and moments from vehicle sensor data, achieving less than **0.05** NRMSE across all wheels.
- Implemented an end-to-end ML pipeline with a FastAPI and Celery-based backend, integrating MLflow and autonomous data drift detection to ensure continuous monitoring while decoupling heavy inference tasks to achieve **sub-50m**s API response times.
- Deployed ML models on Google Cloud's Vertex AI for automated batch inference, dynamically spinning up containers to support **50** concurrent jobs and generating **50,000+** predictions per job within a **15-minute** window.
- Reduced reliance on physical Wheel Force Transducers, delivering **INR 20 Cr** one-time cost savings and **INR 30 lakh** savings per project cycle, while accelerating vehicle validation cycles by **60%**.

Voice-Enabled RAG Application

- Established a scalable database management platform using FastAPI and Postgres to streamline engine testing workflows, improving engineers' productivity by **80%**.
- Architected a voice-enabled RAG pipeline via LangChain, integrating OpenAI's Whisper for Speech-to-Text (STT) conversion to transcribe user audio queries with **94%** accuracy, enabling seamless natural language interactions.
- Designed a high-performance retrieval layer using pgvector, integrating RAG Fusion to generate multiple query variations and Reciprocal Rank Fusion (RRF) to re-rank hybrid search results, achieving a top-k retrieval accuracy above **96.8%**.

Big Data Analytics and RAG Pipeline

- Programmed an end-to-end analytics platform using Apache Spark (PySpark) on Google Cloud Dataproc, automating large-scale vehicle test analysis, improving engineer efficiency by **95%** and delivering **INR 152 Cr** in business impact on product development cost.
- Created a structured RAG system with pgvector to match complex vehicle test parameters, surfacing highly relevant historical data tables and graphs with a **92.8%** Top-K retrieval accuracy.
- Engineered an asynchronous request manager using Celery and Redis to orchestrate long-running analytics jobs, completely eliminating REST API timeouts and enabling parallel execution of up to **1100** concurrent requests.

Leadership & Strategic Initiatives

- Executed advanced analytics on complex enterprise databases across Mahindra Group divisions, uncovering strategic insights that targeted outreach and drove a **10%** increase in customer acquisition for the property business.
- Chaired enterprise-wide AI capability development as a core contributor to the Mahindra AI Academy. Standardized upskilling strategies for the GenAI domain by designing learning paths for **1000+** employees, from entry-level to executives, and accelerated the organizational adoption of AI tools.
- Collaborated with the automobile research division to analyze engine test data and identify testing bottlenecks. Redesigned vehicle testing workflows by optimizing test parameters, resulting in a **15%** increase in test cycle efficiency.
- Directed and mentored a team of 8+ interns through the end-to-end product lifecycle of AI and data projects, ensuring 100% on-time milestone completion while accelerating overall team output.

Data Science Intern- Mahindra Group Oct 2023 – Jun 2024

Call Analytics:

- Improved speech transcription accuracy by **34%** through applying audio preprocessing techniques and integrating Microsoft Azure Speech to Text with Azure AI Translator for multilingual transcription.
- Enhanced sentiment analysis model performance by **6%** through model fine-tuning of DistilBERT on domain-specific transcription datasets.
- Remodeled a highly accurate Named Entity Recognition (NER) system using a custom spaCy transformer pipeline to extract key caller intents and product mentions from audio transcripts, delivering an F1-score of **89%**.
- Visualized call insights and sentiment trends, identifying the **top 5** customer pain points for tailored strategic improvements.

DevOps Intern- MyEra Technologies Apr 2023 – May 2023

- Engineered CI/CD pipelines using GitHub Actions to standardize the build and test workflows of Node.js projects, ensuring **100%** automated test coverage prior to release and reducing deployment failures by **20%**.

EDUCATION

Bachelor of Technology in Computer Science and Engineering (CGPA: 9.48/10) Aug 2020 – Jul 2024 | Jaipur, India

JECRC University

SKILLS

Programming Languages and Core Libraries — Python, C/C++, NumPy, Pandas, SciPy, Polars, JAX, cuDF

Machine Learning/ Deep Learning — TensorFlow, PyTorch, Scikit-Learn, XGBoost, LangChain, LangGraph, OpenAI, LLaMA, Ollama, CrewAI | NLP (Natural Language Processing), Computer Vision, RAG Pipelines, Fine-Tuning (LoRA, QLoRA), Agentic Design Patterns

Cloud, MLOps and DevOps — Google Cloud Platform (Vertex AI), Microsoft Azure (ML Studio), AWS, MLflow, Docker, Kubernetes, CI/CD, Jenkins, Git

Backend Development and Databases — FastAPI, Celery, REST APIs, gRPC, MCP, JSON-RPC, A2A/A2P Protocols, FastMCP | PostgreSQL, MongoDB, Redis, pgvector, Pinecone

Big Data and Visualization — Apache Spark (PySpark), Matplotlib, Seaborn, Plotly

CO-CURRICULAR ACTIVITIES

Winner of Mahindra AI Hackathon- 2024 Oct 2024 – Oct 2024 | India

- Architected an agentic AI system to automate HR (Human Resource) workflows; screened and categorized CVs by role, generated role-specific interview questions, analyzed candidate responses alongside facial expressions, and provided final candidate ratings.